

Generation Interconnection System Impact Study Report

For Queue Project Q20-1490

Lewistown 138 kV

February 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1490. The Interconnection Customer proposes a 144.2 MW solar photovoltaic facility interconnecting at the Lewistown 138 kV substation in the Pennsylvania area. The request was received on June 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 448 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,432,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ft Humbug 345 kV/138 kV TX	Ft Humbug - W Campus 345 KV Line	131.4	356	44.6
Northside Drive - Volcano 138 KV	Ft Humbug - Coronet 500 KV	118.6	180	13.4
Northside Drive - Buxton 138 KV	Coronet - Ft Humbug Substation 500 KV line	117.8	185	13.4
Yeagertown Substation - Buxton 138 KV	Coronet - Ft Humbug 500 KV	108.7	206	13.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ft Humbug 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$21,976,000	49% (\$10,703,000)
Northside Drive - Volcano 138 KV	Rebuild line with higher-capacity conductor	\$5,786,000	64% (\$3,735,000)
Buxton - Northside Drive 138 KV	Rebuild line with higher-capacity conductor	\$6,353,000	64% (\$4,101,000)
Yeagertown Substation - Buxton 138 KV	Reconductor line	\$2,932,000	64% (\$1,893,000)
	Total allocated to project		\$20,432,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.